PORTFOLIO · DATA ANALYST

# Fintech Credit Risk Analytics for BNPL Services

*Identifying high-risk customers and payment behavior patterns in Buy Now, Pay Later transactions*

**Mochamad Rizki Abdul Mughni**

AGENDA

# What's in This Portfolio

| | | |
|---|---|---|
| 01 | **Profile & Experience** | *About me, work experience, and skills* |
| 02 | **Project Overview** | *BNPL dataset, business problem, and pipeline* |
| 03 | **Data Analysis** | *EDA, statistical drivers, and hypothesis testing* |
| 04 | **Customer Segmentation** | *Value, geographic, and demographic segments* |
| 05 | **Deliverable & Impact** | *Power BI dashboard, recommendations, and results* |

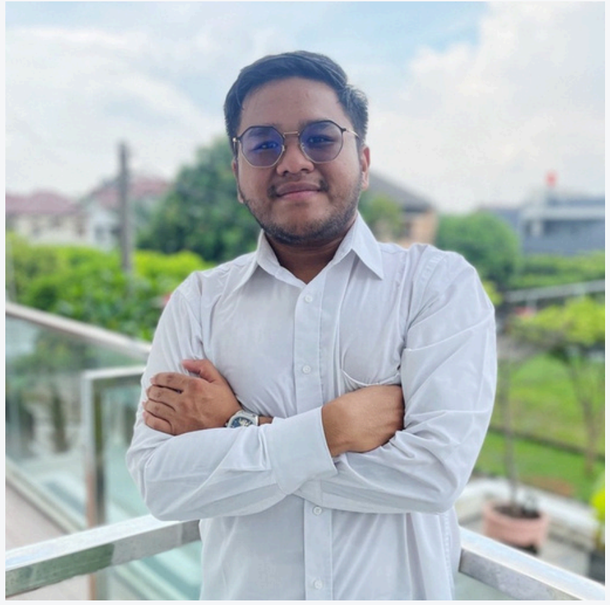
01 · PROFIL

# About Me

Currently pursuing a **Master's degree in Business Analytics** at Universitas Prasetiya Mulya, and holds a **Bachelor's degree in Management** from Universitas Brawijaya — with a strong interest in data analytics and business intelligence.

Currently upskilling through a **Data Analyst Bootcamp at Dibimbing**, and developing hands-on skills in data cleaning, statistical analysis, and data visualization using **Python, SQL, and Excel**, alongside BI tools including **Power BI, Looker Studio, and Tableau**.

Coming from an Human Resources background, now **transitioning into data analytics** combining people-operations experience with technical analysis, as demonstrated in the **BNPL credit risk project** in this portfolio.



# Experience & Organization

### Data Analyst Bootcamp

In Progress

**Dibimbing**

*In Progress · Remote*

- Practicing **SQL** (window functions, CASE-based categorization) and **Python** on analytical case studies, plus inventory management frameworks (EOQ, Q/P systems).

### Human Resources Generalist

**PT Niterra Mobility Indonesia**

*Jan 2026 – Present · 8 mos · Jakarta, Indonesia · On-site*

- Processed and analyzed monthly attendance data for **350+ employees** using Excel/Google Sheets (pivot tables, formulas) for payroll accuracy and HR reporting.
- Supported Compensation & Benefits by assisting Training Needs Analysis (TNA) — data compilation, validation, and summary reports.

### Business Intelligence Analyst — Project-Based Virtual Intern

Most Relevant to This Role

**Bank Muamalat Indonesia × Rakamin Academy**

*Dec 2025 – Jan 2026 · 2 mos · Central Jakarta, Indonesia · Remote*

- Performed data exploration and analysis using **Excel and SQL** on transactional banking datasets to identify sales patterns, trends, and KPIs.
- Designed an **ERD** in MS Access, integrated 4 tables via **MySQL/BigQuery**; built interactive **Looker Studio** dashboards with recommendations.

### Talent Acquisition → PIC Team Talent Acquisition

**upscale.id**

*Dec 2023 – Apr 2024 · 5 mos · Indonesia / Remote · Internship*

- Managed end-to-end recruitment and was promoted to PIC (person-in-charge), onboarding new Talent Acquisition interns.

# Skills & Tools

## Programming

- Python (Pandas, NumPy, Scikit-learn)
- SQL

## Visualization & BI

- Power BI
- Looker Studio
- Tableau
- Matplotlib / Seaborn

## Analytical Methods

- Hypothesis testing (Chi-Square, Mann-Whitney U, Kruskal-Wallis)
- Customer segmentation (RFM-based)
- EDA & correlation analysis

## Soft Skills

- Stakeholder communication — translating HR & BI data into reports for management
- Cross-functional collaboration — recruitment, C&B, and BI project stakeholders
- Business storytelling — turning statistical findings into owner-assigned recommendations
- Adaptability — actively transitioning from HR into data analytics

# Project Overview

**BNPL Credit Risk Analytics & Customer Segmentation**

**Dataset:** Kaggle — Buy Now, Pay Later (BNPL) Fintech ML Dataset

This project analyzes a Buy Now Pay Later (BNPL) dataset of 10,345 customers across 6 countries (2023–2024) to identify customers at risk of default and to segment customers by credit profile and payment behavior.




## Analysis Pipeline

*Data Understanding → Data Cleaning → Feature Engineering → EDA → Hypothesis Testing → Customer Segmentation → Export*

- Identify the key drivers of default among BNPL customers
- Segment customers into Low / Medium / High credit risk
- Recommend credit risk mitigation strategies

BUSINESS CONTEXT

# Background & Business Problem

Buy Now Pay Later (BNPL) lets consumers pay in installments without a credit card. In Indonesia, BNPL has grown rapidly through platforms such as GoPaylater, Kredivo, and Akulaku — but growth has outpaced risk controls.

Without a reliable way to separate risky from safe customers, BNPL providers extend credit limits without weighing debt-to-income ratio, credit score, and repayment history together — driving up default losses.

01

### High Default Rate

**39.05%** of BNPL customers in this dataset defaulted — without segmentation, providers can't tell risky customers from safe ones.

02

### No Early Warning System

**No indicator** flags customers likely to stop paying before losses occur.

03

### Inefficient Credit Allocation

Limits are set without jointly weighing **debt-to-income ratio, credit score, and delay history**.

04

### Business Impact

Defaults raise **cost of risk**; better segmentation can meaningfully lower the **NPL rate**.

DATA UNDERSTANDING

# Dataset at a Glance

**Source**

Kaggle — BNPL Credit Risk

**Raw rows**

10,345 transactions

**Raw columns**

17

**Final columns**

**32** (after feature engineering)

**Period**

2023 – 2024

**Countries**

6 (Germany, UK, Australia, Canada, USA, India)

**Missing / Duplicates**

0 / 0

**Target variable**

default_flag (0/1)

## Key Columns

**user_id** Unique customer identifier

**age** Customer age (18–59)

**employment_type** Salaried / Student / Self-Employed / Unemployed

**monthly_income** Monthly income (USD)

**credit_score** Credit score (300–850)

**purchase_amount** BNPL purchase value (USD)

**bnpl_installments** Installment tenor chosen (3, 6, 9, or 12)

**repayment_delay_days** Days of payment delay

**missed_payments** Number of missed installments

**default_flag** Target: 1 = defaulted, 0 = paid on time

**debt_to_income_ratio** Total debt to income ratio

**risk_score** Customer risk score (0–400)

PIPELINE STAGE 3

# Feature Engineering

*New variables engineered from raw columns to sharpen risk segmentation and enrich the EDA.*

### total_debt_estimate

debt_to_income_ratio × monthly_income estimated USD debt exposure per customer.

### installment_income_ratio

monthly_installment ÷ monthly_income  affordability of the chosen tenor.

### delay_category

Bins repayment_delay_days into No Delay / Mild / Moderate / Severe.

### credit_tier

Buckets credit_score into industry-standard tiers (Poor → Excellent).

### age_group

Bins age into 18–25 / 26–35 / 36–45 / 46–59.

### is_employed / ever_delayed

Binary flags for employment status and any repayment delay.

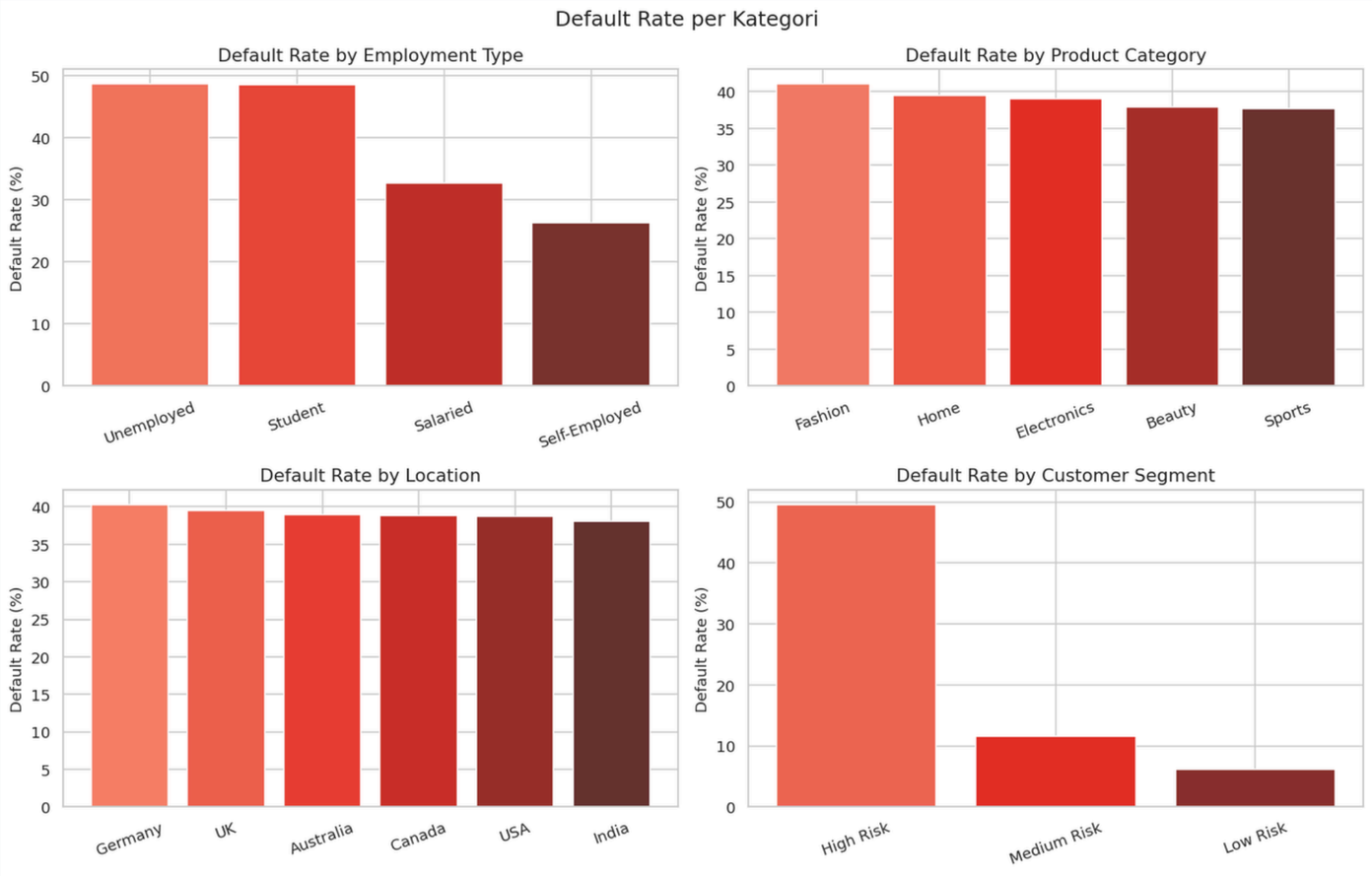
### txn_year / month / quarter / dow

Date components extracted from transaction_date for trend analysis.

### purchase_tier × engagement_tier

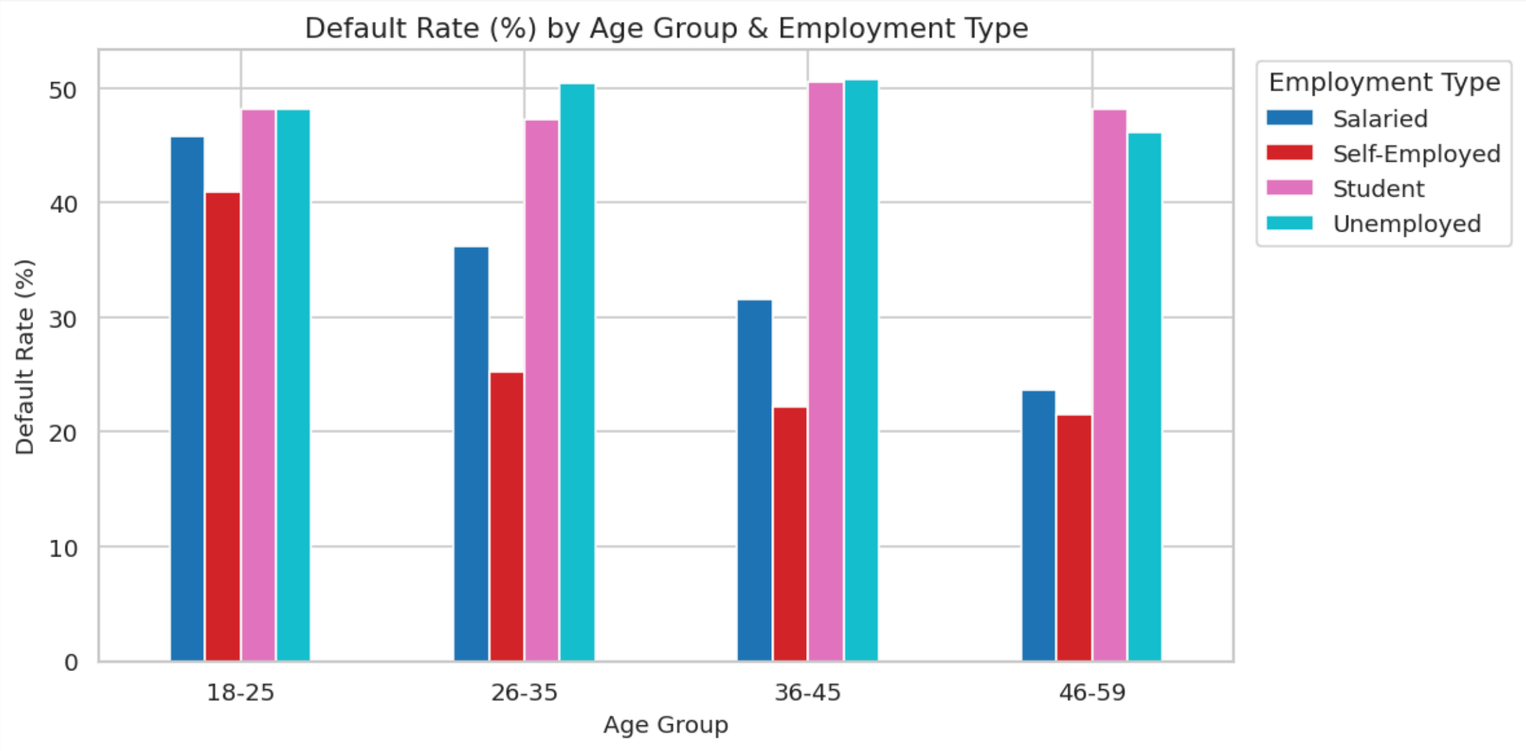
Combined into customer_value_segment  the marketing segmentation used in Section 6.

# Default Rate by Category



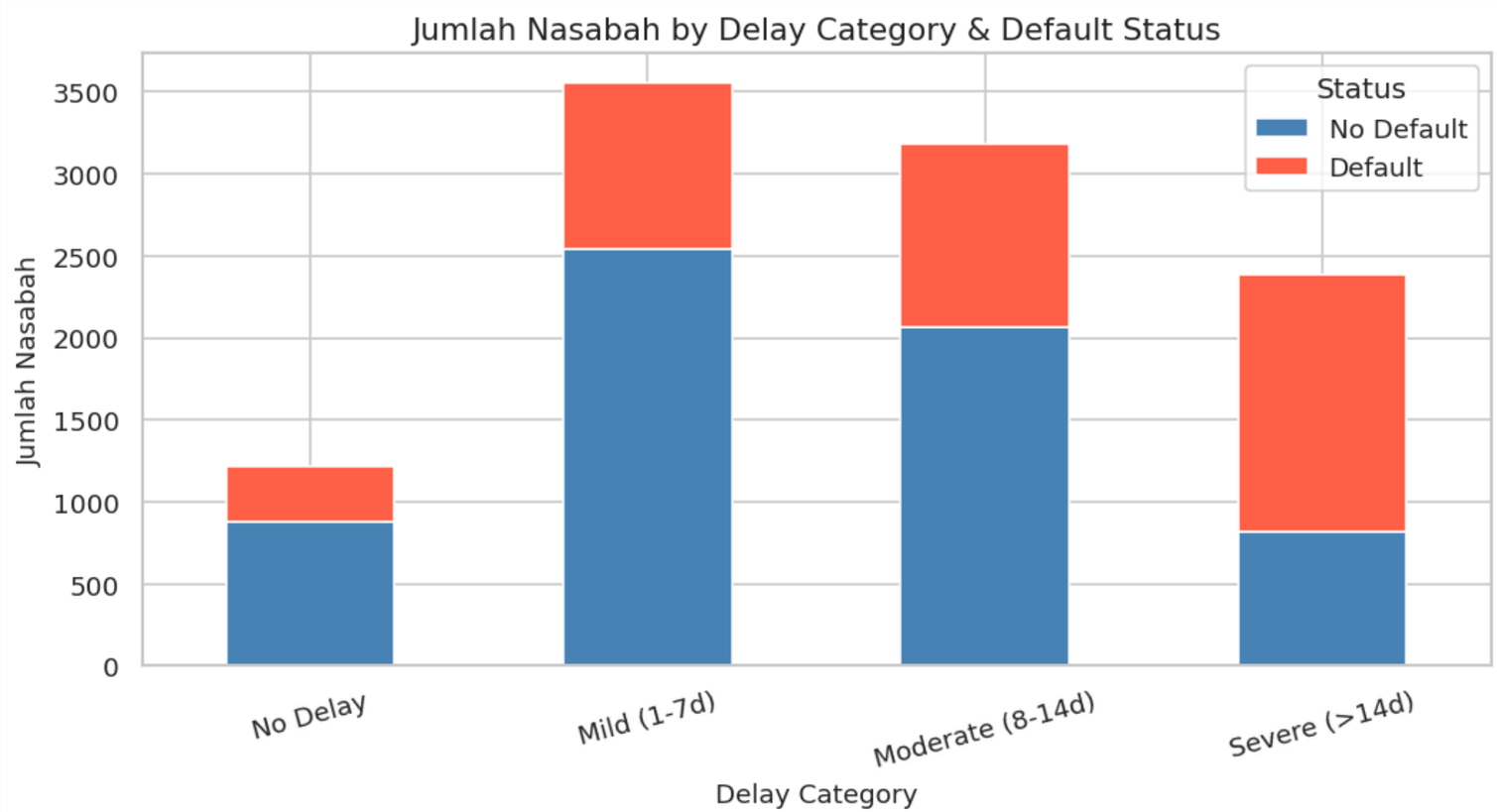

- **Unemployed (48.7%)** and **Student (48.6%)** customers default at nearly **1.9x** the rate of Self-Employed customers (26.2%).
- **Fashion (41.0%)** and Home (39.5%) look like the highest-default product categories, but Chi-Square testing shows this difference is **not statistically significant** (p = 0.168).
- Location shows the same pattern — visible spread but **not significant** (p = 0.830). Employment type and customer segment are the real drivers.
- Default rate by risk segment shows the widest, most significant spread: **High Risk 49.6%** vs **Low Risk 6.1%**.

# Default Rate by Age Group & Employment



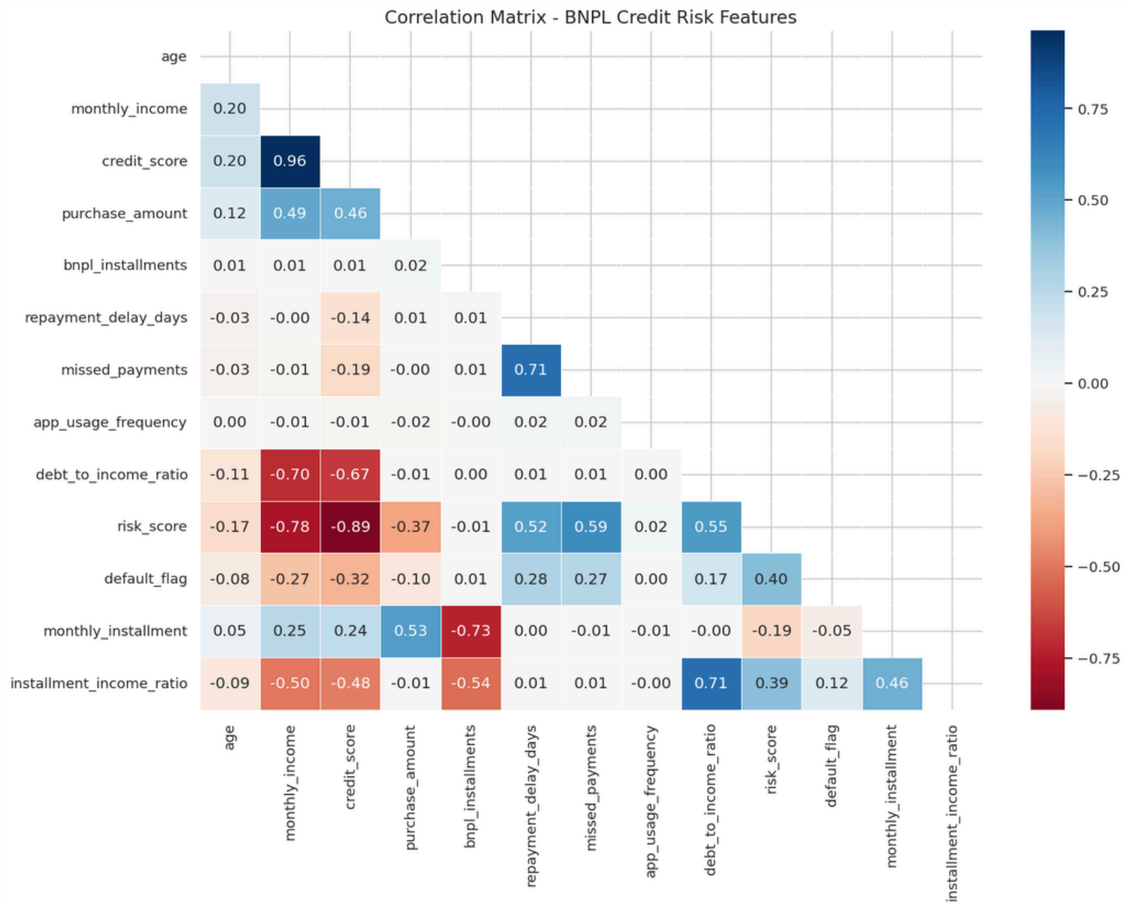

- Among working customers (Salaried, Self-Employed), default rate falls steadily with age — from **~46% (18–25)** to **~22–24% (46–59)**.
- Student and Unemployed customers stay **above 46% at every age**, including ages 36–59 — income stability matters more than age here.
- All employment types show elevated risk at **ages 18–25 (over 41%)**, suggesting younger customers are inherently higher risk.
- **Policy implication:** age-based credit easing should apply only to Salaried/Self-Employed segments, not Student/Unemployed.

# Delay Category & Default Status




- Default rate rises sharply with delay length — from a minority in "No Delay" to the **clear majority in "Severe" (>14 days)**.
- In the Severe category, **1,569 customers defaulted** vs. only **816 who did not** — the critical point of no return.
- The largest customer volume sits in "Mild" delay (1–7 days, **3,557 customers**) — the best window for early intervention.
- **Recommendation:** automate payment reminders at the Mild stage and escalate to restructuring by day 8 (Moderate).

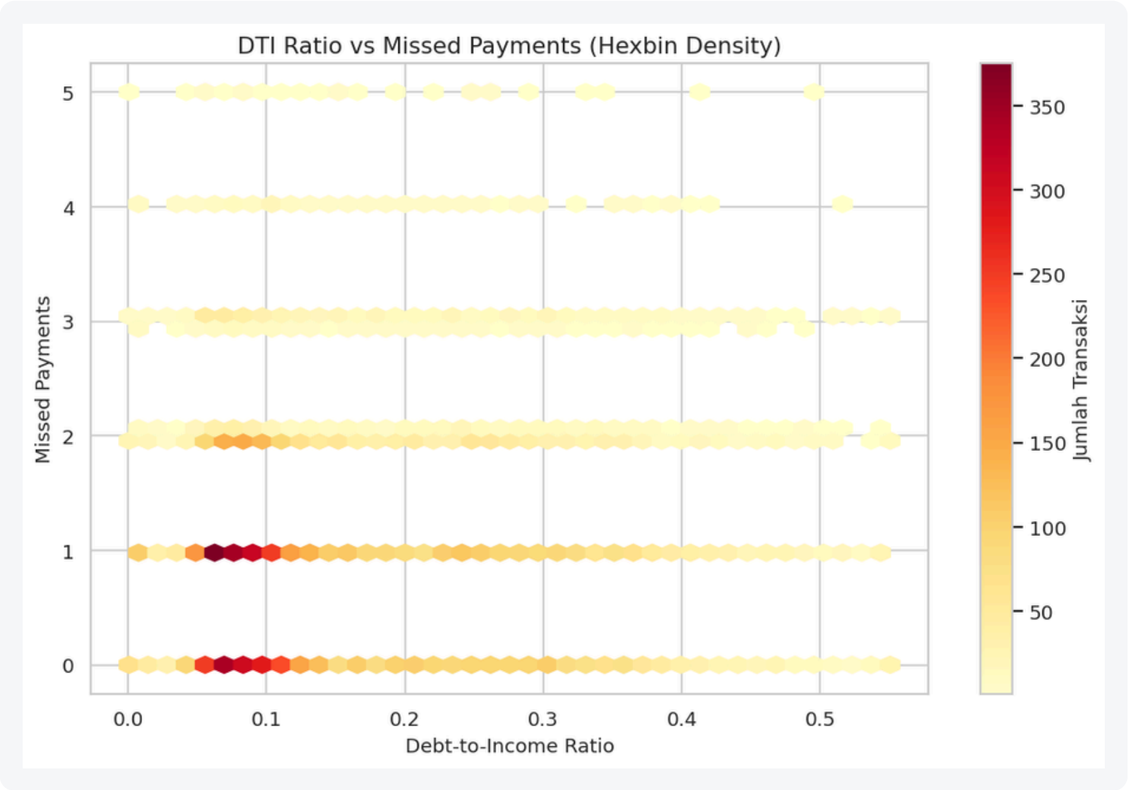
# Multivariate Correlation




- **Credit score (-0.89)** and **monthly income (-0.78)** show the strongest correlation with risk score — this model is **capacity-driven, not behavior-driven**.
- App usage frequency shows **near-zero correlation** with default — digital engagement doesn't predict repayment behavior in this dataset.
- Monthly income and credit score are **nearly collinear (0.96)** — future predictive modeling should retain only one to avoid double-weighting.

# Default vs. Non-Default: Mean Profile

| Feature | No Default | Default |
|---|---|---|
| age | 39.3 | **37.4** |
| monthly_income | $40,969 | **$25,777** |
| credit_score | 483.6 | **392.9** |
| purchase_amount | $4,097 | **$3,796** |
| repayment_delay_days | 7.2 | **11.1** |
| missed_payments | 0.80 | **1.35** |
| debt_to_income_ratio | 0.163 | **0.208** |
| risk_score | 177.0 | **232.2** |




Missed payments and debt-to-income ratio are both low and tightly packed for most customers — high-DTI, high-missed-payment combinations are rare but carry outsized risk.

DATA ANALYSIS

# Statistical Drivers of Default

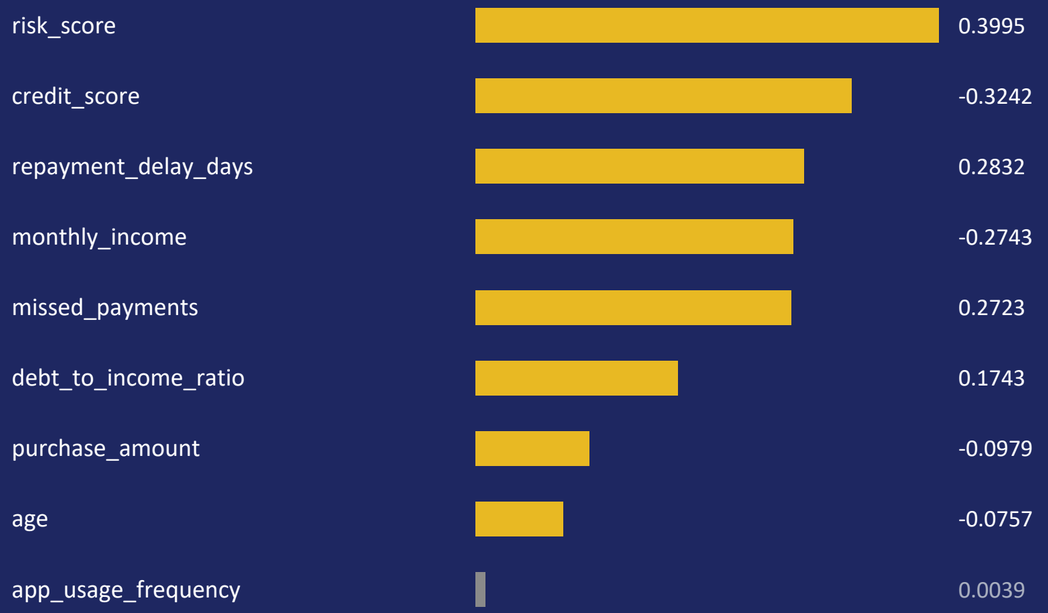
*Formal significance testing confirms which variables actually move default risk, beyond what the charts alone suggest.*

## Chi-Square Test — Categorical Features

| Feature | $\chi^2$ / p-value | Significant? |
|---|---|---|
| customer_segment | $\chi^2$=1313.45, p=< 0.0001 | Yes |
| employment_type | $\chi^2$=418.81, p=< 0.0001 | Yes |
| product_category | $\chi^2$=6.44, p=0.168 | No |
| location | $\chi^2$=2.13, p=0.830 | No |

*Only employment_type and customer_segment significantly relate to default; product_category and location do not.*

## Point-Biserial Correlation — Numeric Features




*risk_score and credit_score are the strongest predictors; app_usage_frequency is not significant (p = 0.69).*

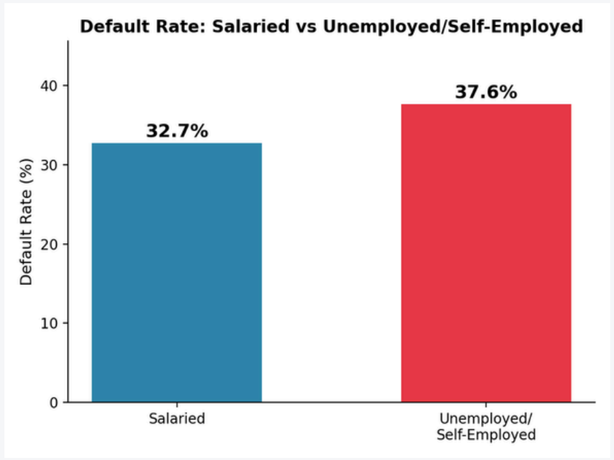
HYPOTHESIS TESTING

# Statistical Tests — H1 & H2

*4 business hypotheses, each tested with normality/variance checks then the appropriate statistical test (α = 0.05).*

## H1 · Employment Status vs. Default

*Chi-Square Test of Independence*

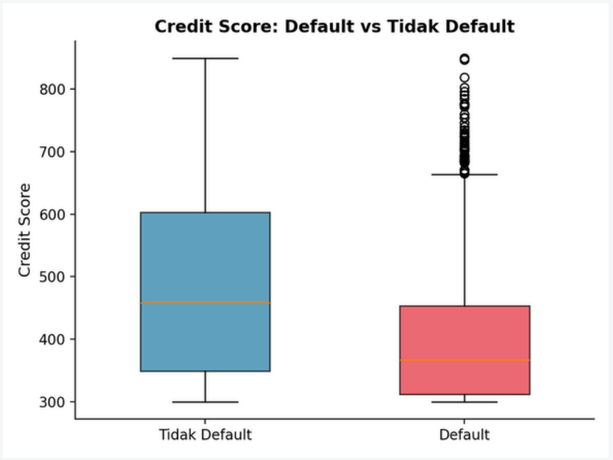



**Reject H0** $\chi^2 = 18.11$, $p = 0.000021$

Customers without steady salary default more (**37.6%**) than Salaried customers (**32.7%**).

## H2 · Credit Score: Default vs. Not

*Mann-Whitney U (non-normal data)*




**Reject H0** U = 17,296,784, $p < 0.000001$

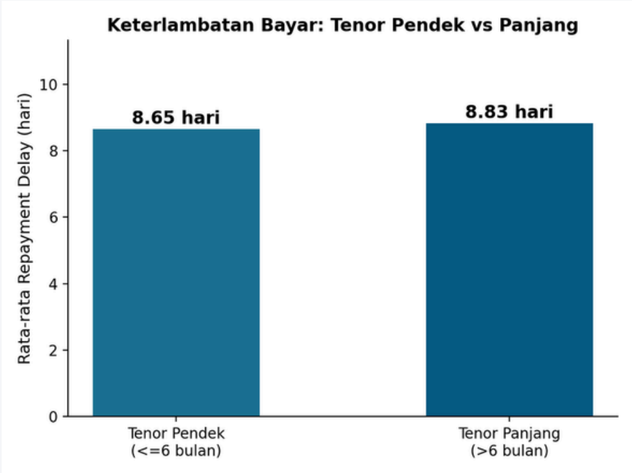
Average credit score is far lower for defaulters (**392.9**) than non-defaulters (**483.6**).

HYPOTHESIS TESTING

# Statistical Tests — H3 & H4

## H3 · Tenor vs. Repayment Delay

*Mann-Whitney U (non-normal data)*

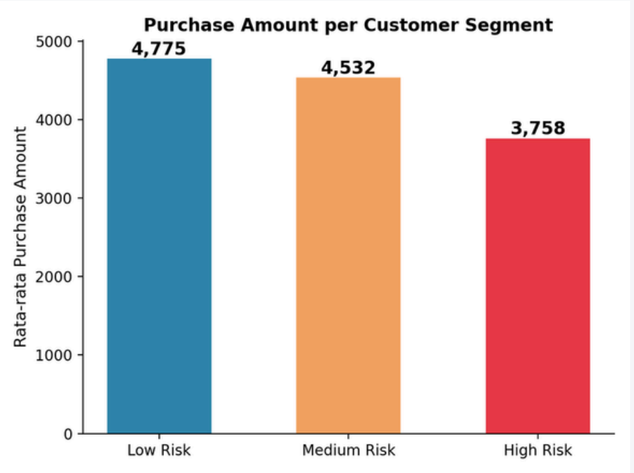



**Fail to reject H0** U = 13,144,072, p = 0.1239

Average delay is nearly identical for short tenor (**8.65 days**) and long tenor (**8.83 days**) — tenor length alone doesn't predict delay.

## H4 · Spending by Risk Segment

*Kruskal-Wallis H (non-normal, unequal variance)*



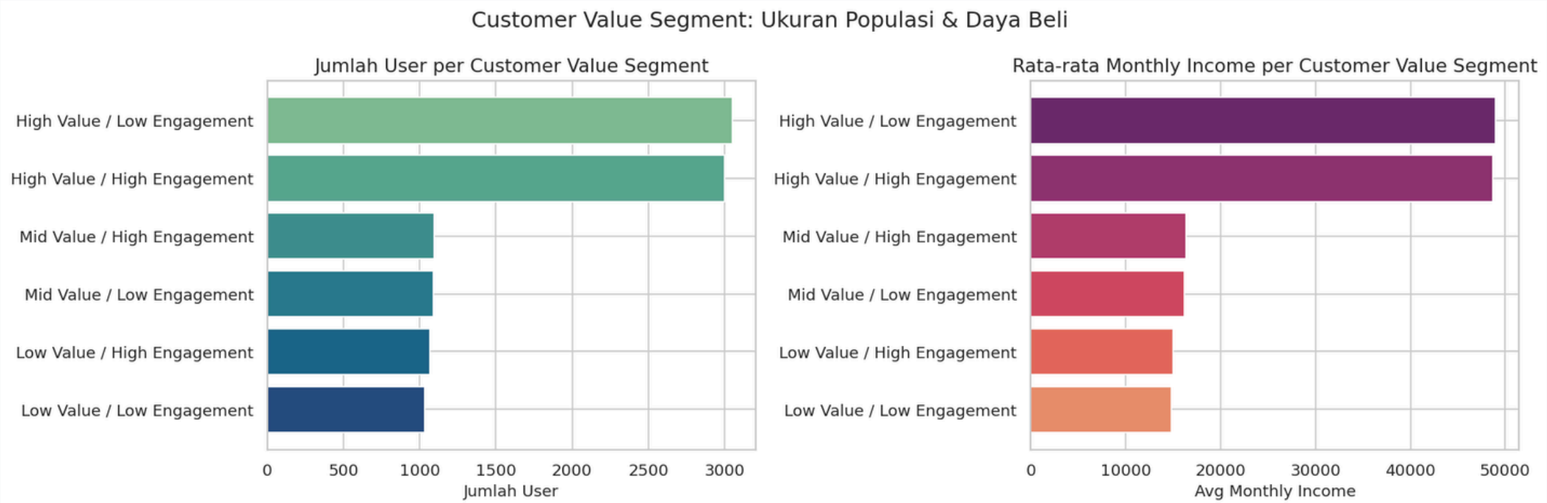

**Reject H0** H = 990.01, p < 0.000001

Counter-intuitively, **Low Risk** customers spend the most on average (**4,775**) vs. **High Risk (3,758)**.

SEGMENTATION

# Customer Value Segmentation

*Built from purchase_amount (value tier) × app_usage_frequency (engagement) — distinct from the dataset's built-in credit-risk segment.*




## Interpretation

- **High Value** segments (Low + High Engagement) make up **58.5%** of all customers — the core revenue base.
- High Value customers earn **~3x more** (avg $48,650–$49,004) than Low Value customers (avg $14,812–$14,980).
- **High Value / Low Engagement (3,051)** are the top re-engagement target: big spenders who rarely open the app.

**29.49%**

**High Value / Low Engagement**

3,051 users · avg purchase $5,000 · avg income $49,004

**29.02%**

**High Value / High Engagement**

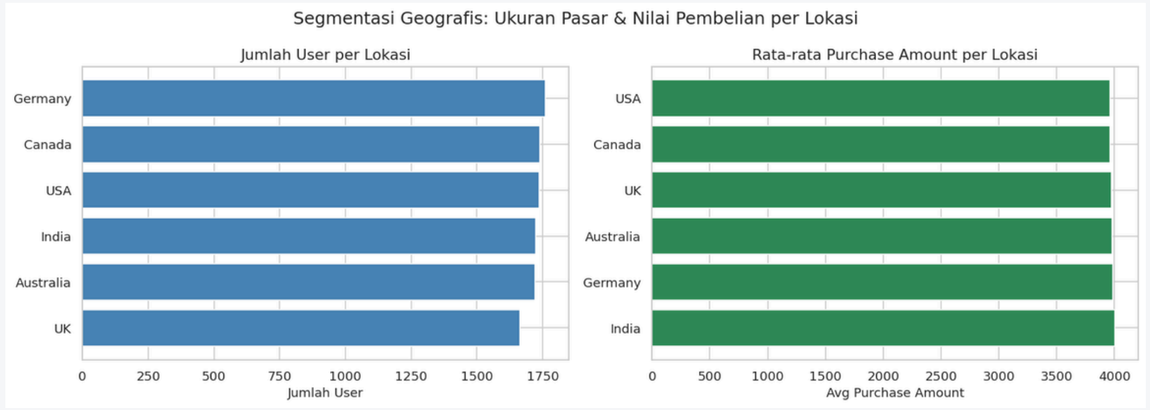
3,002 users · avg purchase $5,000 · avg income $48,650

**6**

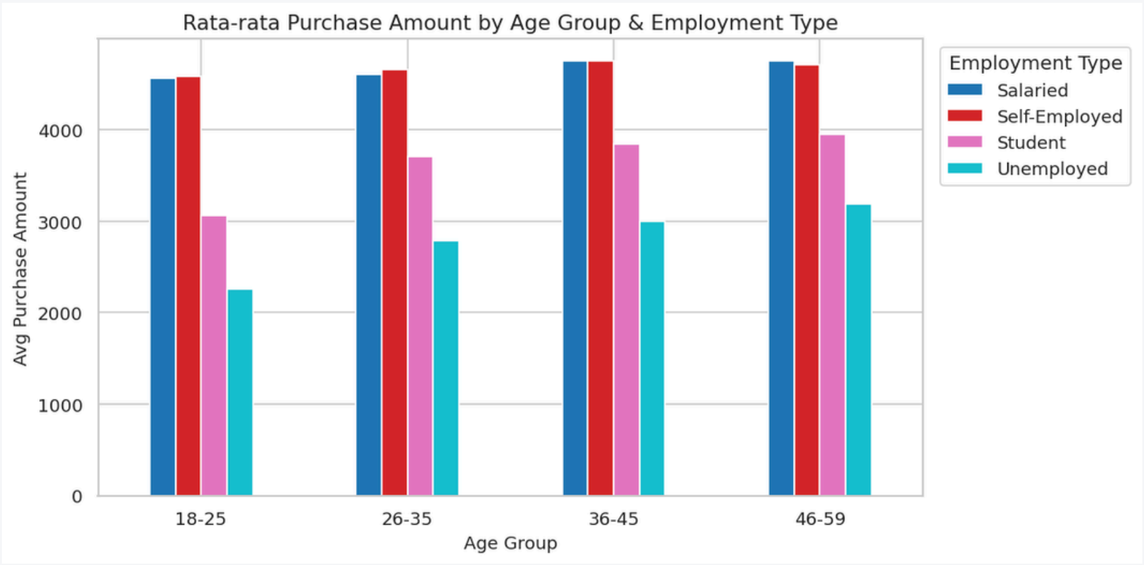
**Total Segments**

Value × Engagement (3 × 2)

# Geographic & Demographic Segmentation




- **Germany** has the largest user base (**1,760**), closely followed by Canada and USA — the six markets are fairly balanced (1,664–1,760 users each).
- **India** has the highest average purchase amount (**$4,005**); differences across countries are modest (~$3,960–$4,005).

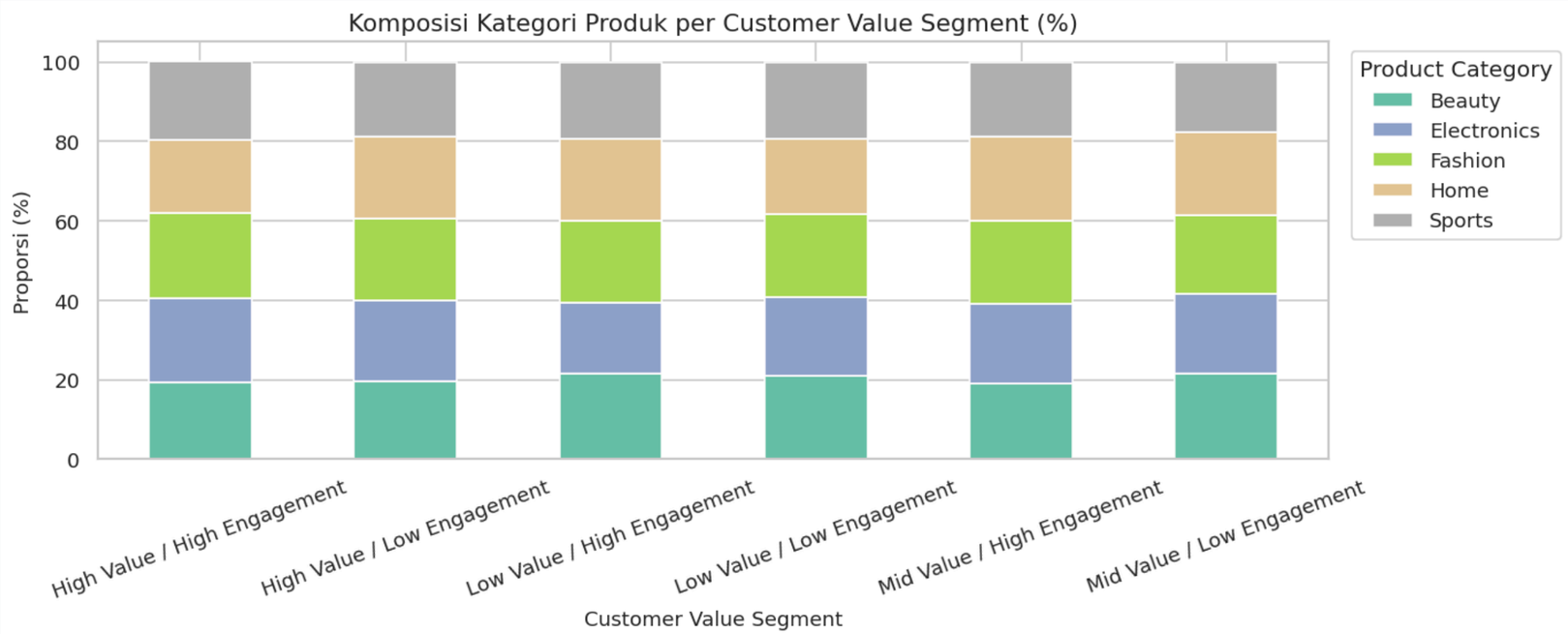



- **Salaried and Self-Employed** customers spend the most across every age group — purchase value tracks income stability, not just age.
- **Students and Unemployed** customers spend noticeably less on average, reinforcing the case for tighter limits on those segments.

SEGMENTATION

# Product Preference by Value Segment

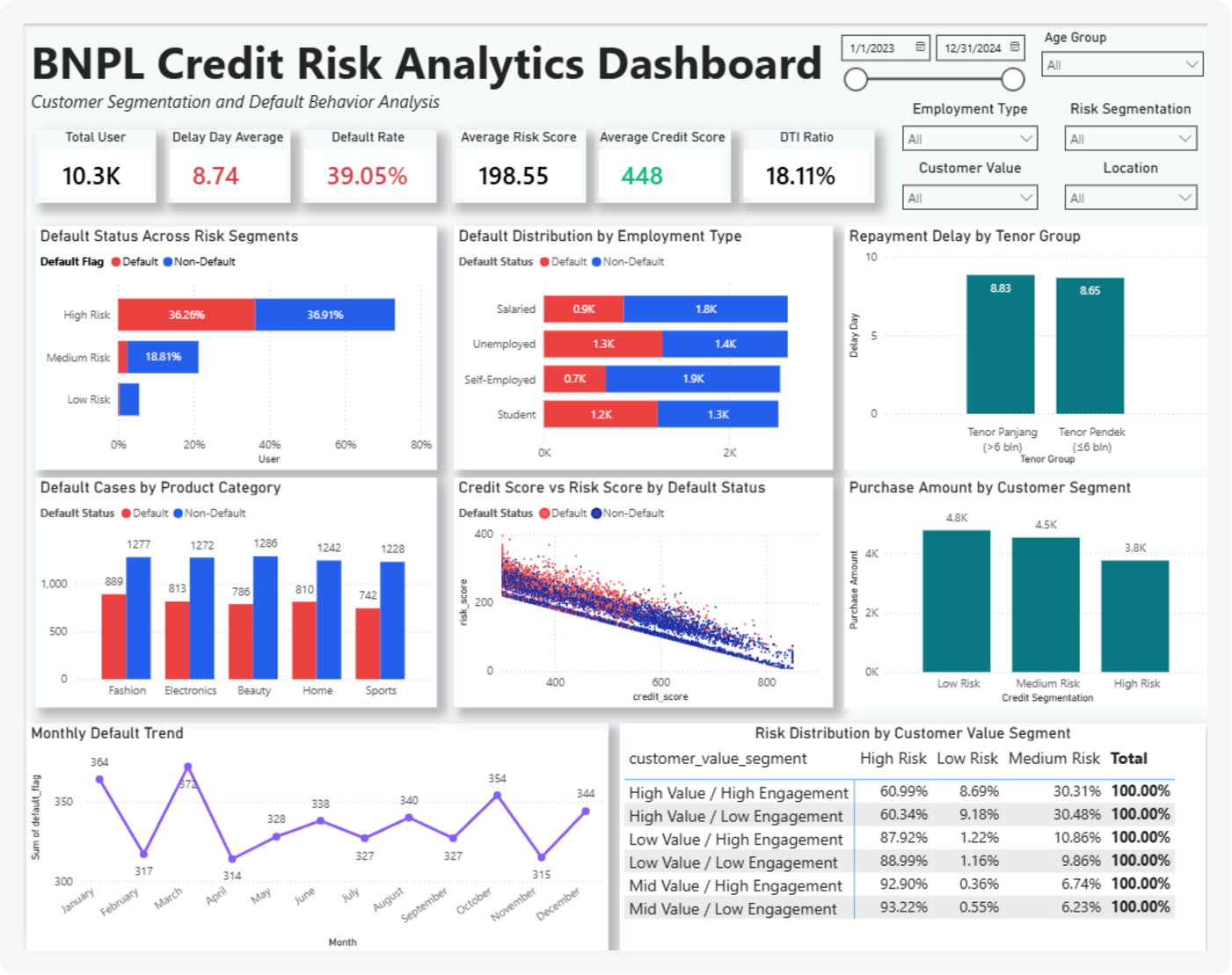
*Product category mix (%) within each customer value segment — a guide for which categories to promote to whom.*




- Product mix is **fairly even** across all six value/engagement segments — no single segment over-indexes heavily on one category.
- This means **cross-category bundling** (rather than category-specific targeting) is the more efficient promotion strategy across segments.

DELIVERABLE

# Power BI Dashboard




**Tools: Microsoft Power BI Desktop**

**KPI Cards**

Total users, default rate, avg. risk & credit score

**Donut Chart**

Customer risk segment distribution

**Line Chart**

Monthly default trend, 2023–2024

**Bar Chart**

Default by employment type

**Scatter Plot**

Credit score vs. risk score

**Slicer Filters**

Location, employment, category, segment

# Recommendations & Actionable Insights

## 01 Dynamic Credit Limit by Risk Segment

*Credit & Risk Management*

**High Risk (73.17% of users):** cap limits at 50% of standard and require strict income verification. **Medium Risk (21.27%):** standard limits with periodic payment monitoring. **Low Risk (5.57%):** eligible for limit upgrades and loyalty programs.

## 02 Early-Warning System on Delay Days

*Collections & Customer Success*

Automated reminders at **day 3** (past the 8.74-day average delay), credit-score-impact notice at **day 7**, and a hard warning at **day 14**. Auto-watchlist any customer delayed past 7 days.

## 03 Strengthen Onboarding for Student & Unemployed

*Onboarding & Underwriting*

Require **proof of alternate income** (bank statements, allowance, family transfers) and a **guarantor requirement** when income doesn't meet the minimum threshold.

## 04 Re-engagement for Low App-Usage Customers

*Product & Marketing*

Shift High Risk promo budget into financial-literacy and payment-reminder campaigns; target cashback incentives at loyal Low/Medium Risk customers, whose average purchase (**4,775 / 4,532**) is well above High Risk (**3,758**).

PROJECT IMPACT

# What This Analysis Delivers

| 10,345 | 3 | 39.05% | 4 |
|---|---|---|---|
| Records Analyzed | Risk Segments Identified | Overall Default Rate | Actionable Recommendations |

**Estimated Debt Exposure Among Defaulters: ≈ $16.1M**

*Derived from total_debt_estimate (debt_to_income_ratio × monthly_income), avg. $3,977/customer × ~4,041 defaulted customers (39.05% of 10,345). A portfolio-level approximation, not a confirmed loss figure — refine with actual write-off/recovery data if available.*

## Relevance to the Role

Coming from an **HR & Talent Acquisition background**, this project is part of an active transition into data analytics — built while completing a **Data Analyst Bootcamp at Dibimbing**. It demonstrates the full analyst workflow expected in the role: **framing a business problem**, cleaning and engineering features from raw data, **validating drivers with formal hypothesis tests**, building customer segments, and **translating findings into concrete, owner-assigned business recommendations**.

# Thank You

*BNPL Credit Risk Analytics & Customer Segmentation*

**Reflection**

*This project pushed me to think in hypotheses, not just charts — testing claims instead of assuming them. If I continue this work, the next step is building a predictive model for individual customer risk scoring, moving from insight to a production-ready tool.*

## Get in Touch


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